Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I think the concept of nested data will help me in this assignment to keep all the data organized and so when I go to access the data for calculating the grade and for the other functions it will be easier. I expect I will be using functions, dataclasses, lists, loops, if statements, conditionals, routes, and states. I want to improve on nesting data while completing my project because it makes the code easier to look through and go back to fix. I also want to work on loops.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think the easiest part of this project will be creating a dataclass for assignments, and adding the assignment grades to a list. This is stuff we have done in previous assignments. The hardest part will be connecting the routes to the pages so that the data stays available as the user moves between pages. Also adding a target grade function may take extra testing and debugging, especially if the score required is above 100 or the list is empty.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think that the grade calculations, routes, and user input will take the most time and effort to complete. First should be, creating a State and Assignment dataclass. Then creating the assignment route that saves names and grades. After that, I will work on the grade calculator function.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Distractions that might interfere with my work would be assignments for other classes, my phone, friends/family and my job. How I will minimize these distractions, is I will work in shorter periods but focus on the project more than other assignments. I will also use a check list to keep myself on track and in order. The resources I will need to use while working will be my course notes, drafter documentation, the program requirements, and the blockpy textbook.

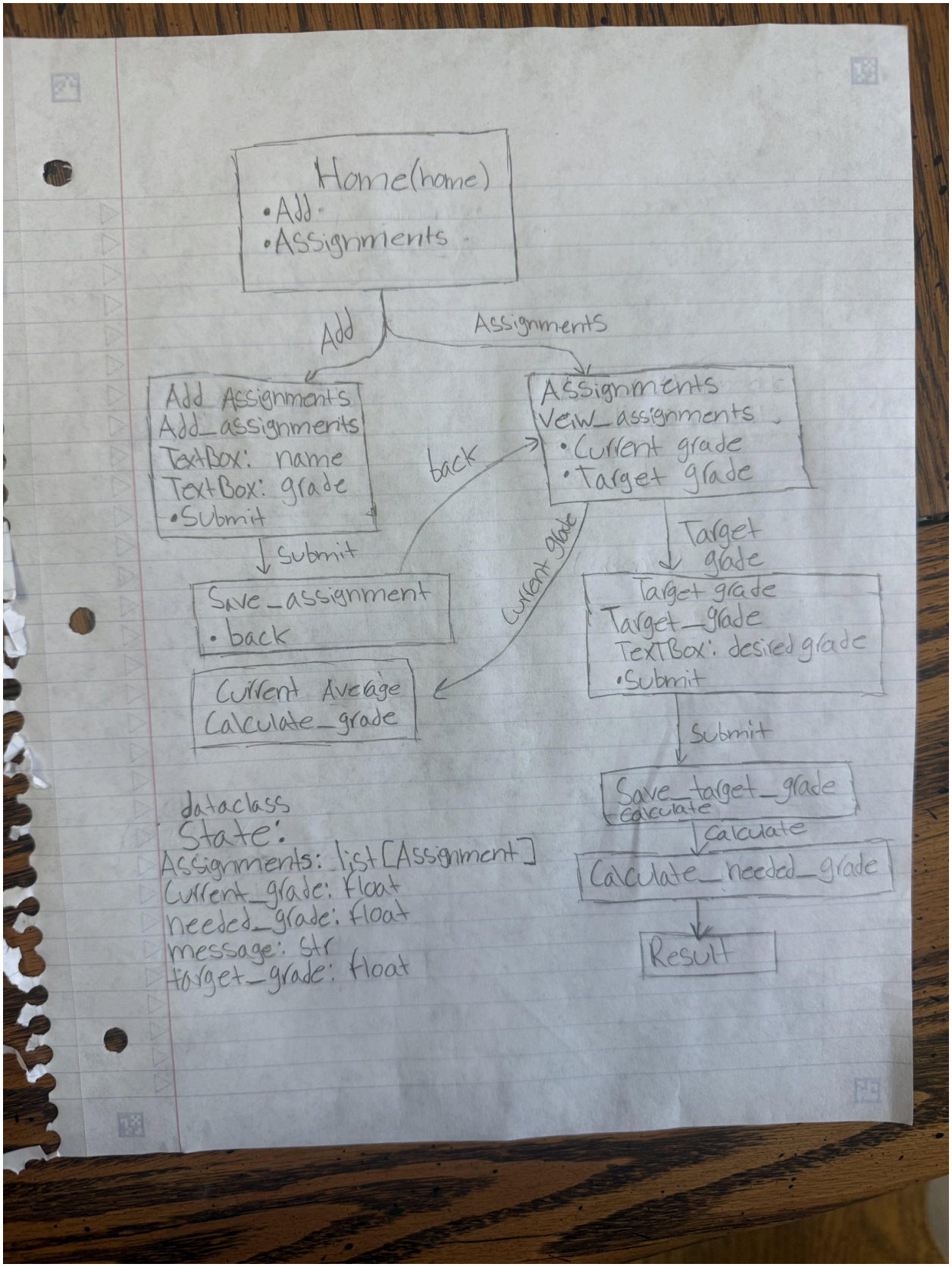
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.



2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	The purpose of my site is to help college students calculate their current grades using their assignment grades from the class easily without having to pay to use all the features. It also helps students see what grade they would need to reach a target grade they want for the class.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	Assignments: list[Assignment] - The list will start empty and then save_assignment appends a valid assignment. Current_grade: float - initial value is 0.0 and calculate_grade updates it using the saved grades. Target_grade: float - initial value is 0.0 and save_target_grade updates it from the user's input. Needed_grade: float - initial value is 0.0 and calculate_needed_grade updates it with required needed grade. Message: str - initial value is "" and routes update it with success, error, or result feedback.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Name: str- the assignment name. Set when assignment is created. Grade: float- students grade as a percent from 0 - 100.
Routes (minimum 7)	Home, Add_assignment, Save_assignment,

The names of your route functions	View_assignment, Calculate_grade, Target_grade, Save_target_grade, Calculate_needed_grade.
Pages (minimum 5) A brief description of the pages in your site	Home- Main page of the website, and takes you to the main features of the calculator. Add Assignments- Lets a student enter an assignment name and grade. Assignments- Displays all assignment names and grades. Current average- Displays the average of all saved assignment grades. Target Grade- Allows the student to enter a desired average and display the grade needed for the next assignment.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	TextBox: assignment name- labels assignment, ex: “work 2”. TextBox: assignment grade- percentage from 0 - 100. TextBox: desired average- wanted average from 0 - 100.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	-when saving an assignment, the if statement will check that the name is not blank, and its grade is between 0 - 100. If invalid, it will not add the assignment and show an error. -when calculating average, the if statement will check if Assignments is empty. If so, it will show a message instead of dividing by 0. - when saving target average, the if statement will

	check that it is between 0 - 100. Otherwise it will reject it and ask for a valid percentage. - when calculating the needed grade, the if statement will check if it is above 100. If so, it will return a message that the target value can not be reached with only one more assignment. -when calculating the needed grade, the if statement will check if it is below 0. If so, it will return a message that the target average has been reached already.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	-Calculate_grade will use a loop to go through every assignment in State.assignments and will add each grade to the total. Then it will divide the total by the number of assignments this will give the average. -View_assignments will use a loop to loop through the list to display each item. This will be able to show the user every Assignment instance they have imputed.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use LLM to help me on this assignment.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.